

Entity-Relationship Diagrams

Five ER Modeling Problems — Solutions and Diagrams (Crow's-Foot Notation)

1. MegaCo Corporation – Divisions, Departments & Employees
2. Airline – Flight
3. FastFlight Airlines – Passenger Reservation System
4. Patty's Playschool – Daycare Registration
5. Bank IT Division – Departments, Employees & Projects

1. MegaCo Corporation

Problem statement: Each of the MegaCo Corporation's divisions is composed of many departments. Each of the departments has many employees assigned to it, but each employee works for only one department. Each department is managed by one employee, and each of these managers can manage only one department at a time.

Entities and Relationships

DIVISION — a division of the corporation.

DEPARTMENT — a department belonging to one division; also has one employee acting as its manager.

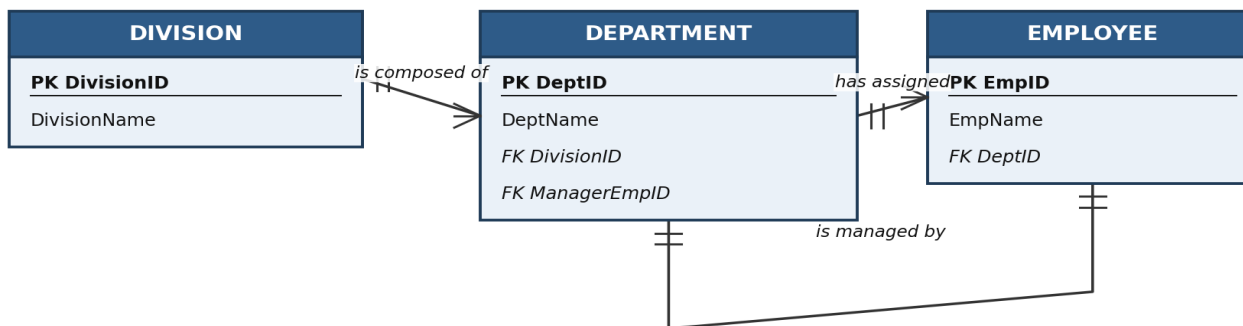
EMPLOYEE — an employee who works for exactly one department; one specific employee may additionally be the manager of that department.

Entity / Attribute	Notes
DIVISION	DivisionID (PK), DivisionName
DEPARTMENT	DeptID (PK), DeptName, DivisionID (FK), ManagerEmpID (FK – references EMPLOYEE)
EMPLOYEE	EmpID (PK), EmpName, DeptID (FK)

Relationships & Cardinalities

- DIVISION (1) —< (M) DEPARTMENT: *"is composed of"* – one division has many departments; each department belongs to exactly one division.
- DEPARTMENT (1) —< (M) EMPLOYEE: *"has assigned"* – one department has many employees; each employee works for only one department.
- DEPARTMENT (1) — (1) EMPLOYEE: *"is managed by"* – one department is managed by one employee, and that employee manages only that one department (1:1).

1. MegaCo Corporation - Division / Department / Employee



2. Airline – Flight

Problem statement: An airline can be assigned to fly many flights, but each flight is flown by only one airline.

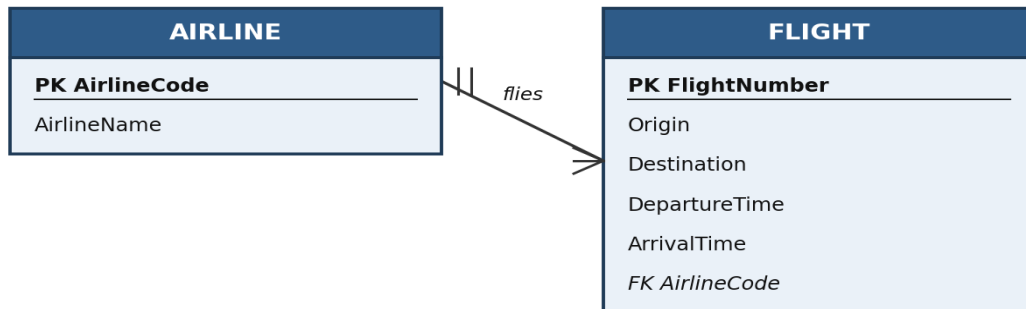
Entities and Relationships

Entity / Attribute	Notes
AIRLINE	AirlineCode (PK), AirlineName
FLIGHT	FlightNumber (PK), Origin, Destination, DepartureTime, ArrivalTime, AirlineCode (FK)

Relationships & Cardinalities

- AIRLINE (1) —< (M) FLIGHT: *"flies"* – one airline operates many flights; each flight is flown by exactly one airline (1:M).

2. Airline - Flight



3. FastFlight Airlines – Reservation System

Problem statement: FastFlight Airlines is computerizing its passenger reservation system. Identified data items: reservation code, flight number, flight date, origin, destination, departure time, arrival time, passenger name, seat number, reservation agent number and reservation agent name. Flight 303, for example, runs every Tuesday and Thursday, leaving Augusta, Maine at 9:23am and arriving in Nashua, New Hampshire at 10:17am. Seat availability is detected automatically.

Entities and Relationships

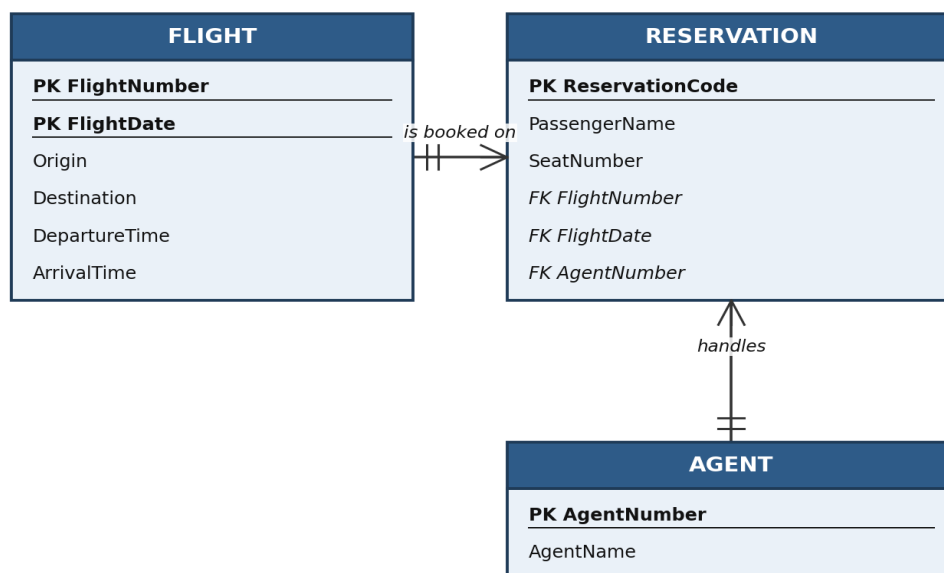
Because a flight number (e.g. 303) runs repeatedly on different calendar dates, the key of **FLIGHT** is the composite (FlightNumber, FlightDate) – this represents one specific day's occurrence of the flight, which is what a passenger actually reserves a seat on.

Entity / Attribute	Notes
FLIGHT	FlightNumber (PK), FlightDate (PK), Origin, Destination, DepartureTime, ArrivalTime
RESERVATION	ReservationCode (PK), PassengerName, SeatNumber, FlightNumber (FK), FlightDate (FK), AgentNumber (FK)
AGENT	AgentNumber (PK), AgentName

Relationships & Cardinalities

- FLIGHT (1) —< (M) RESERVATION: *"is booked on"* – one flight-date can have many reservations (one per seat); each reservation is for exactly one flight-date. Seat availability is derived automatically by comparing reservations for a flight-date against aircraft capacity.
- AGENT (1) —< (M) RESERVATION: *"handles"* – one reservation agent can create many reservations; each reservation is entered by exactly one agent.

3. FastFlight Airlines - Reservation System



4. Patty's Playschool

Problem statement: A parent registers their child or children using a registration form, and may submit more than one form. Each room is assigned an age group (e.g. infants under 1 year, toddlers 1–3 years). A child is assigned to a room based on age and space availability. A room may have one or more employees assigned; an employee is assigned to only one room. The minimum staff needed per room is derived from the number of children and the government child:staff ratio (e.g. 1 employee per 5 infants or 8 toddlers).

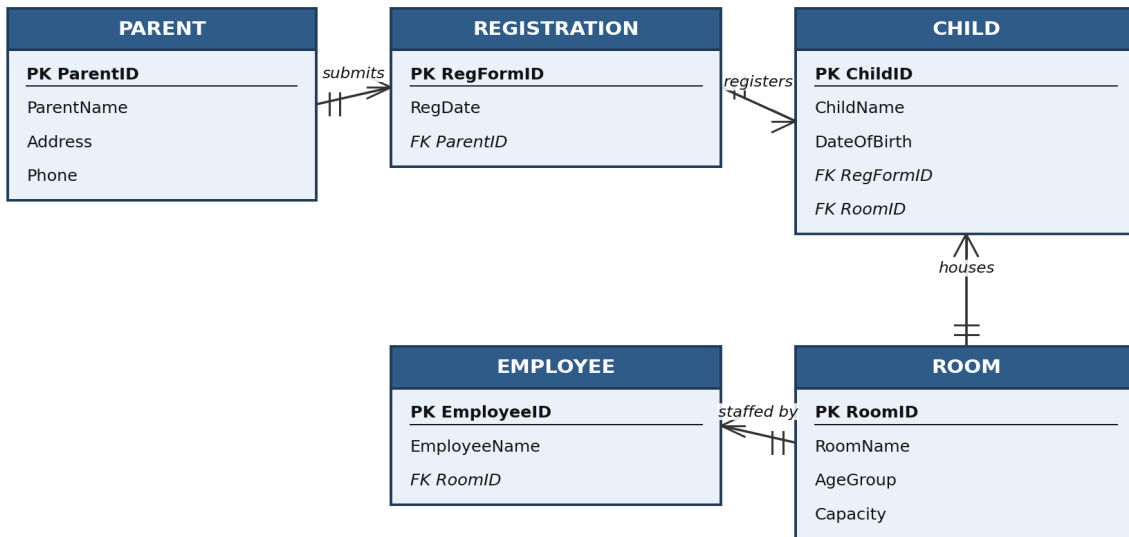
Entities and Relationships

Entity / Attribute	Notes
PARENT	ParentID (PK), ParentName, Address, Phone
REGISTRATION	RegFormID (PK), RegDate, ParentID (FK)
CHILD	ChildID (PK), ChildName, DateOfBirth, RegFormID (FK), RoomID (FK)
ROOM	RoomID (PK), RoomName, AgeGroup, Capacity
EMPLOYEE	EmployeeID (PK), EmployeeName, RoomID (FK)

Relationships & Cardinalities

- PARENT (1) —< (M) REGISTRATION: "*submits*" – a parent may submit many registration forms; each form is submitted by one parent.
- REGISTRATION (1) —< (M) CHILD: "*registers*" – one registration form can register one or more children (siblings); each child is registered on one form.
- ROOM (1) —< (M) CHILD: "*houses*" – a room holds many children (based on age group and capacity); each child is placed in one room.
- ROOM (1) —< (M) EMPLOYEE: "*staffed by*" – a room may have one or more employees assigned; each employee is assigned to only one room. (The government child:staff ratio is a business rule used to compute the minimum required staff for a room, not a stored relationship.)

4. Patty's Playschool



5. Bank IT Division

Problem statement: A large bank operates several divisions; IT is one of them. Within the IT division are many departments, each managed by one manager, and every IT employee belongs to one of these departments. The IT division assigns its employees to one or more ongoing bank projects. A project may be planned but have no employees assigned to it for several months. Each project has a single employee assigned as project leader.

Entities and Relationships

Employees can work on several projects at once, and a project can have several employees – this is a many-to-many relationship, resolved with an associative entity **ASSIGNMENT**.

Entity / Attribute	Notes
DIVISION	DivisionID (PK), DivisionName
DEPARTMENT	DeptID (PK), DeptName, DivisionID (FK), ManagerEmpID (FK)
EMPLOYEE	EmpID (PK), EmpName, DeptID (FK)
PROJECT	ProjectID (PK), ProjectName, LeaderEmpID (FK – references EMPLOYEE)
ASSIGNMENT	EmpID (PK, FK), ProjectID (PK, FK), AssignDate

Relationships & Cardinalities

- DIVISION (1) —< (M) DEPARTMENT: "contains".
- DEPARTMENT (1) —< (M) EMPLOYEE: "employs" – every IT employee belongs to exactly one department.
- DEPARTMENT (1) — (1) EMPLOYEE: "is managed by" – one manager per department (1:1).
- EMPLOYEE (M) —<— (N) PROJECT via ASSIGNMENT: an employee may be assigned to one or more projects, and a project may have zero employees assigned for a period (optional, shown with the hollow circle) up to many employees.
- PROJECT (1) — (1) EMPLOYEE: "is led by" – each project has exactly one employee acting as its project leader.

5. Bank / IT Division - Department / Employee / Project

